

Your grade: 90%

Your latest: 90% • Your highest: 90% • To pass you need at least 80%. We keep your highest score.

Next item →

coach

Almost there! Let's review your answers and then try again.

Your work showed a good understanding of system setup, artificial intelligence applications, and the basics of machine learning and gradient descent. However, there is room for improvement in understanding the components of a Perceptron. Before retrying this assignment, we recommend focusing on this key area:

- **From Perceptron to Neural Network:** In Question 5, your answer included some incorrect components that make up a Perceptron. Try reviewing [this item](#) to refresh your memory on the correct components and their roles in the functioning of a Perceptron.

Good job on the rest of the assignment!

Retry assignment



1. Which command is used to activate a Conda environment in your terminal?

1 / 1 point

- ☐ activate conda [environment_name]
- ☐ conda start [environment_name]
- ☒ conda activate [environment_name]
- ☐ start conda [environment_name]

✔ Correct
Correct! This is the standard command to activate a Conda environment.

2. Which application is an example of artificial intelligence in the present day?

1 / 1 point

- ☒ Personalized recommendations on streaming services
- ☐ Traditional programming loops
- ☐ Manual data entry
- ☐ Static content websites

✔ Correct
Correct! Personalized recommendations on streaming services use AI to suggest content based on user preferences.

3. What differentiates machine learning from classical programming?

1 / 1 point

- ☒ Machine learning models learn from data, while classical programming relies on explicit instructions
- ☐ Machine learning is only used in academic research, while classical programming is used in commercial applications
- ☐ Machine learning models are always more accurate than classical programming
- ☐ Machine learning models do not need any data, while classical programming requires large datasets

✔ Correct
Correct! Machine learning models use data to improve their performance, whereas classical programming relies on predefined rules.

4. Which of the following statements correctly describe the process and importance of gradient descent in training neural networks? Select all that apply.

1 / 1 point

- ☒ The learning rate in gradient descent determines the size of the steps taken towards the minimum of the loss function.

✔ Correct
Correct! The learning rate is crucial in controlling the pace at which the model learns.

- ☒ A very high learning rate in gradient descent can cause the model to overshoot the minimum and fail to converge.

✔ Correct
Correct! A high learning rate can indeed cause the model to diverge instead of converging.

- ☐ Gradient descent guarantees finding the global minimum of the loss function in all cases.

- ☐ Gradient descent is irrelevant in training neural networks and is used only in traditional machine learning algorithms.

- ☒ Gradient descent helps in finding the minimum of the loss function by iteratively updating model parameters.

✔ Correct
Correct! Gradient descent is a key optimization technique for minimizing the loss function during training.

5. Identify the components that make up a Perceptron and contribute to its functioning.

0.5 / 1 point

- ☒ Activation Function

✔ Correct
Correct! The activation function introduces non-linearity to the Perceptron.

- ☒ Bias

✔ Correct
Correct! Bias helps in adjusting the output along with the weighted sum.

- ☒ Weights

✔ Correct
Correct! Weights are crucial in determining the output of a Perceptron.

- ☒ Loss Function

✘ This should not be selected
Incorrect. The loss function evaluates model performance but is not a part of the Perceptron itself.

- ☒ Learning Rate

✘ This should not be selected
Incorrect. The learning rate is a hyperparameter used in training, not a direct component of a Perceptron.

- ☒ Gradient Descent

✘ This should not be selected
Incorrect. Gradient Descent is an optimization algorithm, not a component of a Perceptron.